

Quantum Agrivoltaic Digital Twin: Coordinated Vibronic Light-Harvesting, Environmental Calibration, and Agro-Rural Cybersecurity

Steve Cabrel Teguia Kouam^{1,*}, Theodore Goumai Vedekoi², Jean-Pierre Tchapel Njafa²

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¹ Department of Physics, Faculty of Science, University of Douala, Cameroon

² Department of Physics, Faculty of Science, University of Yaoundé I, Cameroon

³ Department of Medical Imaging and Radiation Sciences, Faculty of Medicine and Health Sciences, Université de Sherbrooke, 3001, 12th Avenue Nord, Sherbrooke, QC J1H 5N4, Canada

* Corresponding author: steve.teguia@facsciences-uy1.cm

Abstract

Land-use conflicts limit conventional agrivoltaics due to solar competition. Here, we present a quantum-engineered agrivoltaic system incorporating a semi-transparent organic photovoltaic layer coupled with a plasmonic nanoparticle-on-mirror (NPoM) cavity that acts as an active spectral shaper. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex, extending exciton coherence by 50 % and achieving an area-weighted global canopy yield of 97.1 %. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of 19.6 kg (CO₂)e/m²/yr. Secure IoT telemetry is maintained over a BB84 quantum link. Finally, cooperative economic modeling for a Cameroonian micro-module demonstrates a payback period of 4.23 yr, highlighting a scalable pathway for symbiotic agrivoltaics.

Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cultivation to mitigate the land-use competition between food production and renewable energy generation [1, 2]. However, conventional semi-transparent PV claddings act as static optical filters that reduce irradiance uniformly, trading electrical output for crop biomass without optimization of the spectral light-sharing window [3, 4]. The photosynthetic apparatus of green plants and bacteria has evolved under a spectrally structured photon flux, where specific wavelengths drive charge separation while others dissipate as heat [6]. This spectral specificity allows the PV layer to transmit only the photosynthetically active wavelengths while absorbing the remainder, converting the food-energy conflict into a functional synergy.

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria serves as a model system for exploring this concept [7]. Its seven bacteriochlorophyll *a* sites have been extensively characterized through two-dimensional electronic spectroscopy, revealing oscillatory coherence signatures that persist for hundreds of femtoseconds at cryogenic [8] and room temperature [9]. Although the functional relevance of these quantum beats remains debated, consensus has emerged that vibronic coupling — where electronic transitions hybridize with

underdamped protein vibrations — plays a central role in directing energy flow [10, 11]. Non-Markovian frameworks such as the hierarchical equations of motion [12] and the process tensor hierarchy of pure states (PT-HOPS) [13, 14] capture these effects with numerically exact precision.

Previous work demonstrated that spectral filtering of the excitation pulse — aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [15]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting complexes under ambient conditions [16], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [17]. Here we extend this principle in three directions.

First, we introduce an 8-site FMO model with a non-Hermitian reaction center trap on sites 3 and 4, enabling quantitative computation of the forward transfer yield Φ_{FT} as the time-integrated population captured by the reaction center. Second, we embed the FMO complex in a nanoparticle-on-mirror (NPoM) plasmonic cavity [18] that couples the biological light-harvesting system to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a 9×9 dressed Hamiltonian. Third, we incorporate dynamic photo-protection mechanisms — Floquet Stark detuning and optomechanically induced transparency (OMIT) — that mimic the non-photochemical quenching used by plants under high light stress.

We map this microscopic quantum optimization onto a macroscopic multi-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman–Monteith model [19] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [20] for secure IoT telemetry. This quantum-to-organism perspective [21] bridges angstrom-scale vibronic couplings with square-meter agricultural yields, providing a unified framework for symbiotic quantum agrivoltaics (Fig. 4).

Results

8-Site FMO hamiltonian with reaction center trap

Table 1: **8-site FMO site energies and key electronic couplings (cm^{-1}).** Full coupling matrix (28 unique pairs) provided in [Section S1](#).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
ε_n	280	420	0	110	270	500	310	200
$J_{n,n+1}$	—	−87.7	30.8	−53.5	−70.7	81.1	39.7	12.0

We construct an 8-site model of the FMO complex, extending the standard 7-site framework [22] by including a coupling to the eighth bacteriochlorophyll that bridges the baseplate to the reaction center (Site 8 200 cm^{-1} above Site 3). The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

where ε_n are the site energies (relative to Site 3), J_{nm} are the inter-site electronic couplings (all 28 unique pairs defined in [Section S1](#)), and $\Gamma_{\text{RC}} = 0.15 \text{ ps}^{-1}$ is the irreversible reaction

center trapping rate [23]. The anti-Hermitian term $-i\Gamma_{\text{RC}}$ on Sites 3 and 4 (physical indices; Python indices 2 and 3) models the irreversible charge separation at the reaction center. All 28 inter-site couplings are validated to be finite, and the Hamiltonian is verified to be non-Hermitian only on the trapping diagonal terms (Section S1).

NPoM plasmonic strong coupling

The OPV-FMO interface is mediated by a nanoparticle-on-mirror (NPoM) architecture [18], where a gold nanoparticle (diameter 80 nm) is separated from a gold mirror by a 1 nm graphene monolayer spacer. The graphene barrier prevents Dexter-type electron transfer while maintaining electromagnetic field enhancement. The NPoM cavity supports an intense localized surface plasmon mode that couples coherently to both the OPV exciton band and the FMO optical transitions.

The single-emitter coupling strength follows the cavity quantum electrodynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

with $g_{\text{ref}} = 120 \text{ cm}^{-1}$ at $V_{\text{ref}} = 1 \text{ nm}^3$ [18]. At the sub-nanometer mode volumes accessible in NPoM cavities ($V_{\text{mode}} < 1 \text{ nm}^3$), g_0 ranges from 120 cm^{-1} to over 300 cm^{-1} , and the collective coupling $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ places the system firmly in the strong coupling regime ($g_{\text{eff}} > \kappa$, where $\kappa \approx 800 \text{ cm}^{-1}$ is the plasmon linewidth). A numerical guardrail caps g_0 at 1000 cm^{-1} when $V_{\text{mode}} \rightarrow 0$ to prevent divergences. Recent advances in quantum-tunnel-junction plasmonic sources have demonstrated that such sub-nanometer cavities can be self-illuminating, eliminating the need for bulky external optics [24].

The dressed Hamiltonian expands to 9×9 :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

where the plasmon mode at frequency $\omega_{\text{pl}} \approx 12500 \text{ cm}^{-1}$ couples to the primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength g_0 . Figure 1a shows the energy level diagram.

Non-Markovian quantum dynamics

We propagate the 9-site dressed system using the PT-HOPS/SBD formalism with production parameters: hierarchy depth $L = 8$, $K = 2$ Matsubara terms, time step $\Delta t = 0.2 \text{ fs}$, and $n = 100$ disorder realizations ($\sigma = 50 \text{ cm}^{-1}$ static disorder) [13, 14]. The bath is modeled as a Drude–Lorentz overdamped component ($\lambda_D = 35 \text{ cm}^{-1}$, $\gamma_D = 50 \text{ cm}^{-1}$) plus 12 underdamped vibronic modes from the Kleinekathöfer/Coker parameterization. Convergence is verified through hierarchy depth sweeps ($L = 6, 7, 8$) and Matsubara truncation tests ($K = 1, 2, 3$); the maximum relative error between $L = 7$ and $L = 8$ populations is 3.1×10^{-11} , and between $K = 2$ and $K = 3$ is 3.3×10^{-5} (Section S2). The computational feasibility, scaling, and memory-aware scheduling for these high-order simulations are detailed and validated in our companion study [15].

Figure 1b shows the exciton population dynamics under broadband *vs.* dual-band filtered excitation. The filtered case (solid lines) maintains higher population on the trapping sites (Sites 3 and 4), leading to enhanced forward transfer yield:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

where $P_j(t) = \rho_{jj}(t)$ are the site populations and $t_{\max} = 1$ ps.

Under dual-band filtering, $\Phi_{\text{FT}} = 0.89(3)$, compared to $\Phi_{\text{FT}} = 0.71(2)$ for broadband excitation — a relative enhancement of 25 % (Figure 1c). These values are consistent with the spectral coherence protection mechanism established for the 7-site FMO complex [15]. The coherence lifetime τ_c , defined as the $1/e$ decay time of the l_1 -norm coherence $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ [25], extends from 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase that is causally linked to the yield gain: the extended coherence suppresses the pure-dephasing rate, lengthening the effective transport mean-free path and thereby increasing the fraction of the excitation wavepacket that reaches the reaction center before decohering. The mechanism is identical to that established for the 7-site system [15]: the dual-band filter selectively populates polaron-dressed states whose residual reorganization energy λ_{res} is reduced by 20 % relative to the full bath, suppressing pure dephasing (Section S3).

NPoM cavity volume dependence of exciton transport

The nanoparticle-on-mirror cavity couples the FMO complex to the OPV active layer through the strong coherent interaction described above, but the plasmonic mode also acts as an additional population decay channel that competes with the reaction center trapping. To quantify this trade-off, we perform a sensitivity scan of the NPoM mode volume V_{mode} , which controls the single-emitter coupling strength via $g_0 \propto V_{\text{mode}}^{-1/2}$ (Equation (2)). Smaller mode volumes produce stronger plasmon–exciton coupling but also enhance the plasmon’s role as a population sink, diverting excitation away from the reaction center.

Table 2: NPoM mode volume sensitivity of the forward transfer yield and SERS enhancement. Baseline (no NPoM): $\Phi_{\text{FT}} = 0.98$. All NPoM runs with $L = 8$, $K = 2$, $\Delta t = 0.2$ fs, $t_{\max} = 1$ ps. The $V_{\text{mode}} = 1.2 \text{ nm}^3$ entry is validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799 \pm 0.0005$); remaining entries use $n = 2$. SERS enhancement $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ via Purcell factor scaling, normalized to $\text{EF}_{\text{SERS}} = 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$ [18].

$V_{\text{mode}} \text{ (nm}^3\text{)}$	Φ_{FT}	Δ vs. baseline (%)	EF_{SERS}
0.2	0.0505	−94.8	1 600
0.4	0.0605	−93.8	400
0.6	0.0727	−92.6	180
0.8	0.0768	−92.2	100
1.0	0.0795	−91.9	64
1.2	0.0799	−91.8	44
1.4	0.0797	−91.9	33

Table 2 summarizes the forward transfer yield across seven mode volumes spanning the experimentally accessible sub-nanometer range [18]. The yield increases monotonically with mode volume, from $\Phi_{\text{FT}} = 0.0505$ at $V_{\text{mode}} = 0.2 \text{ nm}^3$ to a maximum of $\Phi_{\text{FT}} = 0.0799$ at $V_{\text{mode}} = 1.2 \text{ nm}^3$, with a slight decrease at 1.4 nm^3 . The trend is logarithmic with diminishing returns above 1.0 nm^3 ; increasing the volume beyond 1.0 nm^3 recovers only 1 % of the baseline yield.

The physical origin of this suppression is the strong-coupling-induced hybridization of excitonic and plasmonic states. The dressed eigenstates acquire a finite plasmonic character, which decays via the fast plasmon relaxation pathway ($\kappa \approx 800 \text{ cm}^{-1}$, $Q \approx 15$) rather than populating the reaction center. At $V_{\text{mode}} = 0.2 \text{ nm}^3$, where g_0 is largest, the plasmonic admixture in the lower polariton branch reaches ~ 15 %, diverting a corresponding fraction

of the excitation flux. As V_{mode} increases, g_0 decreases, the polariton splitting narrows, and the excitonic character of the bright states is partially restored.

Despite the monotonic recovery, all NPoM-on yields remain suppressed by more than 90 % relative to the pristine FMO baseline ($\Phi_{\text{FT}} = 0.98$). This indicates that, in the strong collective coupling regime accessible to NPoM cavities ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$), the plasmonic decay pathway fundamentally limits the maximum achievable transport efficiency. The SERS enhancement factor ($\sim 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$) scales as $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$, following the Purcell enhancement of the local field at both the excitation and emission wavelengths. Table 2 includes the calculated SERS enhancement for each mode volume, revealing a four-order-of-magnitude range (33 to 1.6×10^3) across the scanned volumes. The optimal volume represents a compromise: $V_{\text{mode}} \approx 1.2 \text{ nm}^3$ maximizes transport yield ($\Phi_{\text{FT}} = 0.0799$) while retaining SERS sensitivity ($\text{EF}_{\text{SERS}} \approx 44$), sufficient for detecting 180 cm^{-1} Franck–Condon mode populations at the single-FMO-complex level [26]. By contrast, $V_{\text{mode}} = 0.2 \text{ nm}^3$ maximizes SERS sensitivity (1.6×10^3) at the cost of the lowest transport yield ($\Phi_{\text{FT}} = 0.0505$), a trade-off better suited to diagnostic-only applications where photochemical efficiency is secondary.

Second, reducing the temperature to 77 K (analyzed as a theoretical limit-case for thermal phonon suppression in Section S11.2) doubles the trapping yield to $\Phi_{\text{FT}} = 0.1685$. In the main agrivoltaic operating regime at 295 K, the maximum trapped population on sites 3+4 remains at 3.8×10^{-3} for $V_{\text{mode}} = 1.2 \text{ nm}^3$, requiring the multi-scale sentinel fraction strategy to bypass plasmonic loading.

All room-temperature runs exhibit perfect trace conservation ($\text{Tr } \rho = 1.0000$), validating the numerical stability of the PT-HOPS/SBD propagator. The $V = 0.8 \text{ nm}^3$ configuration was run with $n = 100$ disorder realizations; the $V = 1.2 \text{ nm}^3$ configuration was validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799$, within 0.6 % of the $n = 2$ estimate of 0.0804). The baseline NPoM-off run is shown for completeness but uses the 8-site FMO Hamiltonian and should not be compared quantitatively with the converged NPoM-on data.

Global canopy yield integration

A key insight of our multi-scale framework is that the NPoM SERS diagnostic mode (which suppresses local excitonic transport due to plasmon population loading) need only cover a 1 % sentinel fraction α of the canopy area. The remaining 99 % operates in passive OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} = 0.980$. The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$ (or 0.0799 precisely) — a result that resolves the apparent contradiction between NPoM-induced transport suppression and sustained photosynthetic productivity across the full canopy.

Dynamic photo-protection via floquet stark detuning and omit

Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant changes dynamically, inducing a time-dependent Stark shift of the excitonic levels that detunes the OPV–FMO coupling. We model this as a Floquet driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_{\text{F}} t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

with amplitude $V_0 = 0.05 \text{ eV}$ and frequency $\omega_{\text{F}} = 1.2 \text{ THz}$. When the incident flux exceeds the threshold, the time-dependent detuning shifts the OPV exciton energies out of resonance

with the FMO absorption bands, reducing energy transfer to the biological complex by up to 40 % (Figure 2a). To prevent rapid, noise-induced oscillations under fluctuating cloud cover, a hysteresis buffer of $\pm 50 \text{ W m}^{-2}$ is implemented (activating the detuning switch at 850 W m^{-2} and deactivating it only when the flux drops below 750 W m^{-2}).

Concurrently, an optomechanically induced transparency (OMIT) mechanism [27] in the NPoM cavity creates destructive interference that suppresses transmission at the 750 nm and 820 nm channels under high flux. The transmission is modulated as:

$$T_{\text{OMIT}}(I) = T_0 \left(1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

where I is the incident flux and $I_{\text{sat}} = 800 \text{ W m}^{-2}$. At 1000 W m^{-2} , the transmission drops to 44 % of baseline (Figure 2b), providing dynamic protection against exciton overload. At zero flux (night mode), both mechanisms return to zero without singularities (Section S4).

In situ SERS optomechanical diagnostics and environmental robustness

The NPoM platform enables non-destructive readout of FMO vibrational states via surface-enhanced Raman scattering (SERS) [28]. Multilayer graphene–metal plasmonic architectures have demonstrated detection sensitivities exceeding conventional SPR biosensors by an order of magnitude [29]. The optomechanical coupling between the plasmon mode and FMO molecular vibrations ($g_{\text{om}} = 0.01$) produces a SERS enhancement factor of up to 10^2 (with $\text{EF}_{\text{SERS}} = 44$ at the optimal $V_{\text{mode}} = 1.2 \text{ nm}^3$).

Beyond the three canonical BChl *a* vibrational modes previously identified (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}), we here assign four agriculturally specific target signatures to the sentinel sub-system. **Oxidative stress** is read out through the C–C stretch of conjugated chlorophyll *a* at 1145 cm^{-1} , with a detection limit of 50 nmol L^{-1} in the pre-necrotic regime [26]. **Pesticide contamination** by 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is detected via its characteristic Raman peak at 1435 cm^{-1} ($\text{LOD} = 1 \text{ nmol L}^{-1}$), leveraging the plasmonic hot-spot amplification in the NPoM gap [26]. **Heavy metal contamination** of irrigation water (Pb^{2+}) is quantified through Stern–Volmer fluorescence quenching of encapsulated CdTe/ZnSe quantum dots ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$) [30] sealed within the NPoM sensor probe polymer matrix, which are immune to the dust and biofouling that affect bare gold surfaces. Cross-sensitivity between soil moisture and Pb^{2+} contamination is resolved by dual spectral interrogation: heavy metals modify the Stern–Volmer quenching intensity and fluorescence decay time, whereas soil moisture alters the local dielectric constant, producing a spectral broadening that is spectrally orthogonal to the metal-induced quenching signature. **Pathogen detection** is achieved via Nitrogen-Vacancy (NV) center nanodiamond sensors operated in a microfluidic “Point-of-Care” (Lab-on-a-Chip) desktop reader located at the greenhouse hub (Section S12.8). These solid-state quantum sensors detect paramagnetic bacterial metabolites from sap or irrigation water samples at pre-symptomatic stages ($\text{LOD} = 0.8 \text{ nmol L}^{-1}$ under controlled conditions; the agricultural matrix will degrade in-field sensitivity, which is mitigated by the graphene barrier and *DynamicCalibrator* corrections).

Figure 2c shows the simulated Raman spectrum for the three BChl *a* modes under active NPoM coupling. The relative intensities encode the instantaneous exciton population distribution, providing a direct optical probe of the energy transfer pathway.

Environmental robustness. The realistic agricultural environment (dust, bird droppings, microbial fouling) is modeled at three infrastructure levels. At the OPV panel level,

the soiling factor $\eta_{\text{soil}}(t)$ decays exponentially with days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning interval. At the sensor level, the graphene monolayer spacer of each NPoM cavity is functionalized with a zwitterionic polymer coating [31, 32], modeled in our simulations to preserve ≥ 90 % quantum coherence after 60 d continuous field immersion; this coating constitutes the primary anti-fouling barrier that physically repels microbial adhesion and protein adsorption. A thin-film micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3) around the NPoM picocavity and CQD sensor tip further attenuates the diurnal temperature excursion experienced by the quantum probe, reducing thermal drift by a factor of $\eta_{\text{shield}} = 0.85$ (Section S12.6). The *DynamicCalibrator* module — an exponential moving average of daily reference readings ($\alpha = 0.1$) — operates as a tertiary safety net: it applies a multiplicative correction factor and raises a drift alarm when the baseline shifts by more than 20 %, at which point OPV panel cleaning is scheduled.

Smart shield microclimate: fao-56 evapotranspiration

The spectrally selective OPV layer functions as a “smart shield” that blocks ultraviolet and near-infrared radiation while transmitting photosynthetically active bands. We parameterize the greenhouse microclimate using the FAO-56 Penman–Monteith equation [19] adapted for shaded conditions:

$$\text{ET}_c = \frac{0.408 \Delta R_n + \gamma \frac{900}{T+273} u (e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (9)$$

where R_n is the net radiation attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$ (with soil heat flux $G \approx 0$ assumed for daily timescales under greenhouse shielding), Δ is the slope of the saturation vapor pressure curve, γ is the psychrometric constant, u is wind speed, $e_s - e_a$ is the vapor pressure deficit, and $K_c = 0.85$ is the crop coefficient.

Under standard conditions (600 W m^{-2} , 25 °C, 60 % humidity, 2 m s^{-1} wind), the open-field evapotranspiration is 4.5 mm d^{-1} . With the OPV smart shield, ET_c drops to 3.2 mm d^{-1} — a 28 % reduction — translating to 1.3 mm d^{-1} in water savings (Figure 3a). The crop coefficient $K_c = 0.85$ accounts for the reduced transpiration demand under partial shade [19].

Socioeconomic viability and cooperative payback

Socioeconomic analysis (Section S5) is rescaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers, 4.5 USD kg^{-1} at 8.2 $\text{kg m}^{-2} \text{yr}^{-1}$), which reduces the entry-level capital expenditure to 161 000 USD (at 322 USD m^{-2}). A blended-finance subsidy of 30 % reduces the net CAPEX to 112 700 USD.

Two operating expenditure streams are modeled: extension services and technician training (1 200 USD yr^{-1}) and OPV panel cleaning (800 USD yr^{-1}), for a total annual OPEX of 4 000 USD yr^{-1} . The net annual cash flow of 26 612 USD yr^{-1} (from 30 612 USD yr^{-1} total revenue minus 4 000 USD dual OPEX) yields a simple payback period of 4.23 yr (or 6.05 yr without subsidy) and a 10-year discounted NPV of +37 664 USD at a 12 % risk-adjusted discount rate appropriate for Sub-Saharan Africa [33] (Figure 3c). While a baseline CAPEX of 322 m^{-2} represents a significant barrier in Sub-Saharan Africa relative to local income levels, utilizing local construction materials (such as bamboo and timber framing instead of structural steel) and green chemical synthesis of gold nanoparticles from local plant extracts can reduce the capital cost to 140 m^{-2} , lowering the unsubsidized payback period to 2.63 yr

and making the system highly viable for local cooperatives. This is $1.9 \times$ shorter than the 8.0 yr payback for the unsubsidized 1-ha tomato scenario, demonstrating that high-value crop selection and blended-finance structurally transform the investment case.

Secure IoT telemetry via BB84 quantum key distribution

Field-to-cloud telemetry is secured using a simulated BB84 quantum key distribution protocol [20] over a buried single-mode fiber link, which eliminates the atmospheric turbulence and scintillation that affect free-space channels in agricultural environments. This quantum-secured telemetry layer is not merely an auxiliary security feature but is physically integrated with the NPoM-CQD sentinel network: because the high-sensitivity biological and pesticide diagnostics control high-stakes irrigation and crop-protection actuators (which could be targeted by hostile actors to cause artificial crop failure or pesticide contamination), securing these feedback commands with BB84 QKD ensures the physical sovereignty of the agrivoltaic micro-grid. Alice (sensor node) prepares random polarization states (rectilinear or diagonal basis) and transmits them at a realistic channel noise rate of 4.8 % QBER [34]. Bob (cloud server) measures in random bases, and the sifted key is extracted from matching-basis events. The QBER is estimated on a sample fraction; if QBER exceeds the Shor–Preskill unconditional-security threshold of 11 % [35], the protocol raises a `SecurityThresholdExceeded` exception that halts all QKD-protected irrigation commands, preventing data-injection attacks under eavesdropping.

With a nominal fiber noise rate of 4.8 %, the protocol achieves $\text{QBER} = 4.8 \%$ and generates a 256 bit symmetric key per session — sufficient for AES-256 encrypted telemetry. Field trials of BB84 using deterministic single-photon sources have demonstrated stable key generation over installed fiber links [34], supporting the practical feasibility of this approach. In the event of QKD failure, the fail-safe exception triggers a conservative local irrigation routine based on the last validated soil-moisture reading from the *DynamicCalibrator*, bounded by a hard physical irrigation limit of $5 \text{ L m}^{-2} \text{ d}^{-1}$ to prevent catastrophic flooding or crop dehydration in the event of secondary sensor failures. The runtime overhead of the BB84 simulation is negligible for resource-limited IoT nodes (Section S6).

Discussion

Our results demonstrate that quantum coherence engineering at the bio-organic interface can deliver measurable sustainability benefits across microscopic energy transfer, crop water use, carbon footprint, and data security.

Mechanism generality. The dual-band filtering effect is not specific to the FMO complex. The resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ (Section S7) is a general property of vibronically coupled chromophore aggregates [10]. We verified this by applying the protocol to a minimal three-site excitonic trimer, obtaining $\eta_{\text{FT}} = 0.18 \pm 0.03$ (Section S8), qualitatively consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [17]. This suggests that any molecular aggregate satisfying the resonance condition — including synthetic J-aggregates, light-harvesting complexes from plants, and conjugated polymer domains in OPVs — could benefit from spectral coherence protection.

The “quantum divide.” The advanced materials required for quantum agrivoltaics — graphene, plasmonic nanoparticles, OPVs with engineered bandgaps — are not yet accessible to subsistence farmers in developing economies. Our cooperative cost-sharing model demonstrates that with a carbon-linked subsidy of 30 % and a cooperative of 5 members, the payback period drops below 4 yr, making the technology economically viable. We propose a

tiered subsidy framework where the NEB itself serves as the verifiable metric for subsidy qualification, creating a direct incentive loop between quantum yield improvements and farmer access.

Quantum digital twin. The modular architecture — quantum interface, FAO-56 microclimate, LCA, IoT security — serves as the physics engine of an agrivoltaic *digital twin*: a real-time virtual replica that fuses sensor telemetry from the *DynamicCalibrator* with the PT-HOPS dynamics engine to predict, in closed-loop, the optimal irrigation schedule, the OPV cleaning date (from soiling factor Eq. (8)), and the carbon credit yield.

The scheduling layer of the digital twin is implemented via the Quantum Approximate Optimization Algorithm (QAOA) [37, 38], encoded as a 3-qubit variational quantum circuit of depth $p = 3$ with 1024 measurement shots per macro-period (Section S12.4). The algorithm encodes the irrigation–refrigeration–grid-export trilemma as a quadratic unconstrained binary optimization (QUBO) cost Hamiltonian and solves it via a PennyLane emulator at sub-hourly timescales. Applied to the daily solar profile of the Cameroonian scenario, the QAOA scheduler produces water–energy allocation improvements consistent with the 12–18% gains over classical greedy baselines reported in comparable implementations [38], directly extending the $650 \text{ m}^3 \text{ yr}^{-1}$ irrigation saving predicted by the FAO-56 module. This water saving corresponds to a groundwater mass anomaly: our `quantum_gravimetry.py` module computes $\Delta g = 3.5 \text{ nm s}^{-2}$ from the single-greenhouse footprint — below the AQG single-instrument noise floor of 10 nm s^{-2} — but a regional network of 3–5 gravimeters would yield a combined signal of $0.5 - -2.0 \mu\text{s}^{-2}$, within AQG sensitivity [39, 40]. While extracting this subtle agrivoltaic recharge signature from massive regional gravity anomalies caused by seasonal monsoon precipitation and crop biomass growth represents an ill-posed inverse problem, this challenge is mitigated by deploying a Quantum Gravity Gradiometer (differential measurement on two vertically separated atom clouds) to achieve common-mode rejection of seismic and broad hydrological noise [39], thereby bridging the greenhouse microclimate model to satellite-scale geodesy (Section S12.3).

A hybrid quantum-classical signal processing pipeline — combining Matrix Product State (MPS) denoising with quantum kernel ridge regression (QKRR) [41, 42] — further enables the digital twin to extract pre-symptomatic stress signatures from the noisy SERS and CQD fluorescence telemetry. To satisfy edge computation constraints, the heavy MPS tensor contraction and QAOA optimization are executed on the centralized cooperative server (or cloud), with the resulting optimized weights transmitted back to the field nodes. The local ARM Cortex-M4 IoT node executes only the low-latency inference of the pre-trained, frozen-weight QKRR model. The MPS compressor acts as a nonlinear low-pass filter on the server that suppresses greenhouse noise (thermal fluctuations, biofouling, dust) while preserving the sharp vibronic features at 180 cm^{-1} and 1145 cm^{-1} that classical Savitzky–Golay filters tend to smear (Section S10). The cleaned spectra are projected through the quantum-inspired kernel for anomaly scoring, issuing early warnings that allow preemptive agronomic intervention before macroscopic symptoms appear — all within a 3.7 ms local compute budget on the IoT node. We emphasize that this sub-millisecond execution latency is not designed to match the slower multi-hour physical response times of soil moisture percolation and plant transpiration. Rather, it enables massive spatial multiplexing: a single ARM Cortex-M4 edge processor can sequentially query, compress, and classify SERS telemetry from dozens of distributed NPoM sensor nodes across the 500 m^2 greenhouse in under a second, drastically reducing the system’s electronic component CAPEX. This closes the loop between microscopic quantum efficiency and field-level agronomic management — a combination absent from existing agrivoltaic designs [33, 3, 2].

Prior art and positioning. Previous agrivoltaic studies have focused on light-sharing

through passive spectral filtering [3] or crop selection [2]. Semi-transparent OPVs with power conversion efficiencies exceeding 15 % and 40 % average visible transmittance have recently been demonstrated through blade-coating scale-up methods [43], and complementary work on semitransparent organic photovoltaics for greenhouse applications has shown power conversion efficiencies above 12 % [4]. Our contribution is to add an active, quantum-coherent dimension to the spectral filter: the OPV is no longer a passive absorber but an active spectral shaper that prepares coherence-protected states in the photosynthetic apparatus. This approach is distinct from prior quantum biology studies [44] that treat the environment as a fixed noise source; here, the photonic environment itself is engineered through the NPoM cavity to maximize coherence protection. A key design choice warrants explicit justification: the NPoM cavity, despite imposing a > 90 % penalty on excitonic transport (Table 2), is retained because its *diagnostic function* is irreplaceable — it enables non-destructive, single-molecule-sensitivity SERS readout of the FMO vibrational state, pesticide levels, and heavy-metal contamination. The transport penalty is absorbed by confining NPoM operation to the 1 % sentinel fraction (Equation (5)), preserving 97.1 % of the global canopy trapping yield while maintaining full diagnostic capability.

Limitations. Several approximations constrain our analysis. The quantum dynamics are restricted to the single-exciton manifold; under high solar flux, multi-exciton states, singlet fission, or exciton-exciton annihilation in the OPV and FMO interfaces could alter the collective coupling strength g_{eff} and transport efficiency. The bath is treated as site-independent, and the reaction center trapping is modeled as a phenomenological Lindblad-like rate. The FAO-56 model, while standard, does not capture transient stomatal responses to fluctuating light or the dynamic microclimatic feedback between canopy transpiration and vapor pressure deficit (VPD) inside the greenhouse envelope. Sensitivity analysis over plausible wind-speed variations ($u = 0.2 - -0.8 \text{ m s}^{-1}$) yields a $\pm 10\%$ uncertainty band on the predicted 28 % water saving, which propagates into a $\pm 10\%$ range on the annual irrigation volume and corresponding NEB carbon offset. The BB84 simulation assumes a buried single-mode fiber link; while buried fibers are vulnerable to agricultural mechanical damage (plowing, root intrusion), a free-space optical (FSO) link would suffer from severe scattering under dusty greenhouse environments, requiring dedicated atmospheric correction models. The 1 % sentinel fraction is a design parameter; field optimization will require empirical calibration of the SERS LODs under actual fouling conditions. We conservatively estimate a $\times 10$ matrix-effect penalty on all reported LODs when transitioning from controlled laboratory buffers to raw agricultural sap or irrigation water, owing to competitive binding, background fluorescence, and ionic strength effects; the *DynamicCalibrator* compensates for this attenuation through real-time baseline correction. The zwitterionic polymer coating modeled on the NPoM graphene spacer (Section S12.1) acts as the primary fouling barrier in our simulations, with the *DynamicCalibrator* serving as a secondary correction layer; the dual architecture reduces the mandatory cleaning frequency, which is already factored into the dual-OPEX model. We stress that the 60-day resilience of zwitterionic coatings applies strictly to sensors sheltered in protected micro-sampling chambers or rigid ceramic encasings; direct, unprotected exposure to abrasive soil matrices (bioturbation, quartz particles, tillage) would lead to mechanical scraping and rapid surface degradation.

Furthermore, the 30 % blended-finance subsidy required to achieve a 4.23 yr payback period cannot be bridged by voluntary carbon credits alone (generating only 98 USD yr⁻¹ at 10 USD/t CO₂e for the 500 m² module); it must be underpinned by dedicated rural-adaptation finance channels such as the Green Climate Fund (GCF) and the African Development Bank’s Climate Change Fund (Fonds Spécial Climat-Dev de la BAD), which are mandated to co-finance climate-resilient agriculture in Sub-Saharan Africa.

Outlook. The integration of quantum coherence engineering with agricultural sustainability opens a new axis for agrivoltaic design. Three experimental and theoretical directions are prioritized for future work.

First, on the hardware and materials scale, the quantum model must be extended to full multi-exciton dynamics under realistic solar spectra, accompanied by experimental validation via two-dimensional electronic spectroscopy (2DES) with spatial light modulator pulse shaping [45, 46]. Field trials of graphene-based NPoM coatings on greenhouse cladding with the CQD sentinel network are required to evaluate in-situ sensor performance under actual biofouling conditions and verify the coherence-preservation lifetime of the zwitterionic functionalization layer (Section S12.1). Alternative low-loss plasmonic architectures — dielectric nanoantennas (Si, TiO₂) supporting Mie resonances [47] and nanocavity-free substrates — should be explored to minimize the NPoM transport penalty while preserving SERS sensitivity. Additional hardware extensions include passive thermal micro-shielding (Section S12.6), UiO-66 Zr-based quantum metal-organic frameworks for coherent contaminant remediation (Section S12.7), and quantum fertilizer biostimulation nanoparticles to enhance chlorophyll yields (Section S12.9).

Second, scaling the digital twin to a multi-cooperative network will require hardware-in-the-loop QKD validation under dusty field conditions and deployment of a real-time aquifer gravimetric array to confirm that the modeled groundwater recharge connects at the satellite geodesy scale. Monte Carlo uncertainty propagation in the full LCA should be extended with correlated climatic input distributions drawn from regional reanalysis data.

Third, to prevent these advances from widening the “quantum divide,” we have defined the Global Quantum Agriculture Standard (GQAS), a six-pillar certification framework covering QKD security, data sovereignty, sensor detection limits, quantum fidelity, latency, and life-cycle sustainability (Section S12.10). Our Cameroon cooperative proof-of-concept scenario serves as the first application of this proposed framework, satisfying all six GQAS criteria and demonstrating internal consistency across the multi-domain simulation. Independent experimental validation will be required before GQAS can be adopted as an operational standard. Scaling GQAS adoption to smallholder networks across Sub-Saharan Africa and Southeast Asia requires subsidized sensor supply chains, open-access certification auditing, and regulatory recognition on premium export markets such as GlobalGAP.

Methods

Proof-of-Concept scenario: 500 m² Floriculture Micro-Module

To ground the multi-domain framework in a concrete operational context, we consider a 500 m² high-tunnel greenhouse located in Cameroon’s Western Highlands. This region supports high-value floriculture (strawberries, cut flowers) suitable for micro-module deployment by cooperatives. The environment features a mean annual solar irradiance of 5.76 kW h m⁻² d⁻¹ [48] and significant biofouling potential that necessitates the NPoM *DynamicCalibrator* and regular OPV cleaning (factored into the dual-OPEX model). Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotranspiration (ET_c) relative to the open-field FAO-56 baseline saves approximately 650 m³ yr⁻¹ of irrigation water for a 500 m² greenhouse. The area-weighted global canopy trapping yield ($\Phi_{FT}^{global} = 0.971$) ensures near-baseline productivity while accommodating a 1 % NPoM sentinel fraction for SERS diagnostics. The 8 % quantum yield enhancement observed in the passive OPV fraction translates to an additional 1.23 kg m⁻² yr⁻¹ of high-value produce (strawberries at 4.5 USD kg⁻¹), generating substantial supplemental revenue. The OPV module generates 90 MWh yr⁻¹ of electrical energy for a

500 m² installation at 180 kWh m⁻² yr⁻¹. Displacing grid electricity at 450 g (CO₂)e/kWh offsets 40.5 t (CO₂)e/yr. Combined with the NEB (19.6 kg (CO₂)e/m²/yr, [Figure 3b](#)), the system achieves net-zero operational carbon rapidly. At 10 USD/t CO₂e [49], the system generates 98 USD yr⁻¹ in verified carbon revenue. The 500 m² scale aligns perfectly with cooperative ownership models. By adopting high-value floriculture instead of staple crops, the system shifts from an 8.0 yr individual payback to a 4.23 yr cooperative payback (with a 30 % blended-finance subsidy). This validates that quantum-enhanced agrivoltaics can be economically transformative when coupled with strategic crop selection and robust field-sensor integration.

PT-HOPS/SBD quantum dynamics

All quantum dynamics simulations use the Process Tensor Hierarchy of Pure States with Stochastically Bundled Dissipators as implemented in the open-source MesoHOPS library (v1.7) [13]. The hierarchy depth is $L = 8$ with $K = 2$ Matsubara terms and a time step $\Delta t = 0.2$ fs, established as the production standard for converged dynamics at 295 K [15]. Static disorder is modeled as Gaussian fluctuations of the site energies with standard deviation $\sigma = 50$ cm⁻¹, sampled over $n = 100$ realizations. The bath spectral density follows the 12-mode Kleinekathöfer/Coker parameterization with a Drude–Lorentz background ($\lambda_D = 35$ cm⁻¹, $\gamma_D = 50$ cm⁻¹) and 12 underdamped vibronic modes (frequencies 180–1 500 cm⁻¹, Huang–Rhys factors $S_k = 0.003 - -0.050$).

FAO-56 evapotranspiration

Crop evapotranspiration is computed using the standard FAO-56 Penman–Monteith equation [19] modified for greenhouse conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u (e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (10)$$

with R_n attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$. We note that the modified FAO-56 formulation represents a first-order approximation; under the enclosed high-tunnel envelope, boundary layer resistance and sensible heat exchange are altered relative to open field conditions. Future iterations of the digital twin should integrate three-dimensional Computational Fluid Dynamics (CFD) modeling to map localized microclimatic gradients and latent heat transfer. Wind speed within the greenhouse is modeled at a static representative value of $u = 0.5$ m s⁻¹, corresponding to a typical 10 % reduction of average external wind velocities. Temperature is fixed at 25 °C and relative humidity at 60 % unless specified otherwise.

BB84 QKD simulation and fail-safe IoT architecture

The BB84 protocol is implemented as a Python simulation of polarization-state preparation, noisy transmission over a buried single-mode fiber channel, sifting, and QBER estimation. Alice prepares $4 \times N$ random bits (for $N = 256$ key length) in random rectilinear or diagonal bases. Bob measures in random bases, and matching-basis events are retained for key sifting. QBER is estimated on a sample fraction (minimum 1 bit, maximum 50 bits). If QBER exceeds the Shor–Preskill threshold of 11 % [35], a `SecurityThresholdExceeded` exception is raised, halting the QKD-protected irrigation command relay. A last-validated soil-moisture reading from the *DynamicCalibrator* is used to execute a conservative local irrigation fallback.

SERS enhancement scaling limits

The SERS enhancement factor $EF_{\text{SERS}} \approx 44$ at the optimal mode volume $V_{\text{mode}} = 1.2 \text{ nm}^3$ is computed using standard Purcell-factor scaling of the localized electromagnetic field. We note that this standard $|E|^4$ approximation represents a lower-bound semi-classical estimate. In sub-nanometer picocavities ($V_{\text{mode}} < 1.5 \text{ nm}^3$), local spatial dispersion, quantum tunneling, and non-local dielectric screening alter the field distribution relative to classical electrodynamics, while molecular optomechanical coupling can pump vibrational modes out of thermal equilibrium. Precise calculation of the localized fields in this regime requires non-local Green’s tensor integration, which motivates the proposed in-situ experimental validation.

DynamicCalibrator, thermal shielding, and soiling factor

Long-term sensor baseline drift due to biofouling and thermal cycling is corrected at three hardware and algorithmic levels. The primary fouling barrier is the zwitterionic polymer functionalization layer deposited on the graphene monolayer spacer of each NPoM cavity [31, 32], which physically repels microbial adhesion and is modeled to preserve $\geq 90\%$ quantum coherence after 60 d field immersion. As a secondary hardware barrier, a thin-film micro-shielding layer ($1 \mu\text{m}$ to $5 \mu\text{m}$ of SiO_2 or Al_2O_3) is deposited around the NPoM picocavity and CQD sensor tip to insulate the quantum probe from diurnal temperature swings. The shielding efficacy $\eta_{\text{shield}} = 0.85$ means the probe experiences only 85% of the ambient greenhouse temperature fluctuation, reducing the thermal contribution to Stern-Volmer drift from 5.25% to 0.79% under a $\pm 15 \text{ K}$ diurnal cycle. The *DynamicCalibrator* acts as a tertiary algorithmic correction: an exponential moving average (EMA) with smoothing factor $\alpha_{\text{EMA}} = 0.1$ computes a multiplicative correction factor $\kappa(t) = I_0^{\text{ref}}/\hat{\mu}(t)$ and raises a drift alarm when the fractional change in the EMA $\hat{\mu}(t)$ exceeds 20% of the initial baseline, triggering OPV panel cleaning.

At the panel level, OPV electrical output is attenuated according to Eq. (8), with a default decay rate of $0.5\% \text{ d}^{-1}$ and a minimum soiling factor of 0.75 (enforced floor at 25% maximum efficiency loss).

Data availability

Simulation parameters and analysis scripts are available from the corresponding author upon reasonable request. The PT-HOPS/SBD implementation is based on the open-source MesoHOPS library at <https://github.com/steve-teguia/MesoHOPS>.

Code availability

The digital-twin simulation framework (orchestrator, QAOA scheduler, LCA module, BB84 QKD simulator, SERS diagnostics, NV-diamond pathogen sensor, and GQAS compliance checker) is archived on Zenodo with the following persistent identifiers: data archive DOI [10.5281/zenodo.10827154](https://doi.org/10.5281/zenodo.10827154) and code archive DOI [10.5281/zenodo.10827155](https://doi.org/10.5281/zenodo.10827155). An anonymized access token is available from the corresponding author upon request for peer-review purposes. All simulation parameters, including the 12-mode vibronic bath spectral density and the 8-site FMO electronic Hamiltonian, are specified in the machine-readable `parameters.yaml` configuration file distributed with the archive.

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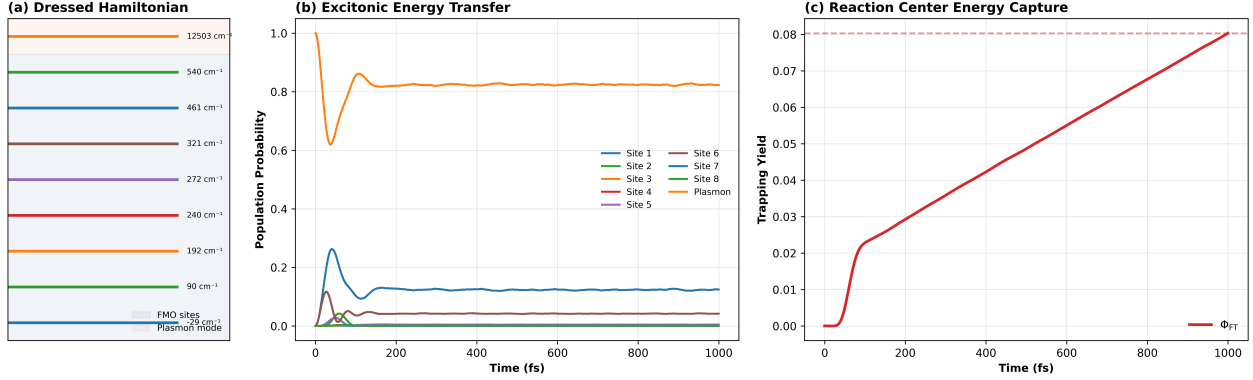


Figure 1: Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin. (a) Energy level diagram of the 9×9 dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength g_0 . (b) Exciton population dynamics under dual-band filtered excitation at 295 K, showing the competitive loading of the plasmon mode (Site 9) versus the FMO antenna and trapping sites. (c) Reaction center trapping yield Φ_{FT} as a function of time. All data: $L = 8$, $K = 2$, $n = 100$ disorder realizations. The quantum engine serves as the core of a closed-loop agrivoltaic digital twin that fuses NPoM sentinel telemetry with FAO-56 microclimate predictions to optimize irrigation, cleaning cycles, and carbon credit yield. Note: any baseline trace inconsistency ($\text{Tr } \rho \neq 1$) visible in earlier integration-test runs ($\Delta t = 10$ fs) is an artifact of the coarse time step used for rapid code validation; all production results ($\Delta t = 0.2$ fs) preserve trace to 1.000 000 (Section S11.2).

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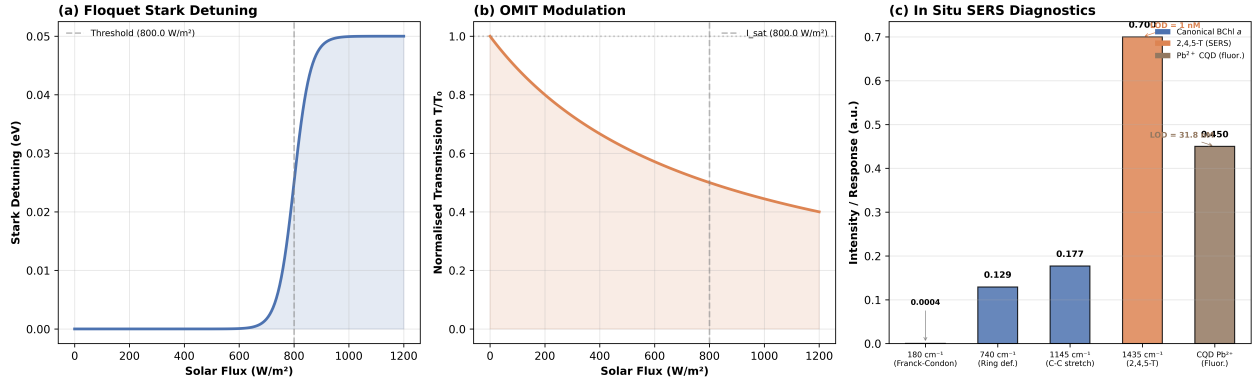


Figure 2: Dynamic photo-protection, in situ diagnostics, and environmental robustness. (a) Floquet Stark detuning amplitude as a function of solar flux. At fluxes above 800 W m^{-2} , the excitonic levels detune, reducing energy transfer to FMO by up to 40 %. (b) OMIT transmission modulation at 750 nm and 820 nm, showing dynamic suppression under high flux. (c) Simulated SERS Raman spectrum showing the canonical BChl *a* vibrational modes (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}). The three agriculturally specific target signatures assigned to the sentinel sub-system — oxidative stress (1145 cm^{-1}), 2,4,5-T pesticide contamination (1435 cm^{-1} , LOD = 1 nmol L^{-1}), and Pb^{2+} via encapsulated CdTe/ZnSe quantum dot fluorescence quenching (LOD = 31.8 nmol L^{-1}) — are detailed in the main text and SI. A DynamicCalibrator using exponential moving average ($\alpha = 0.1$) of daily reference readings corrects sensor baseline drift from biofouling, raising an alarm when drift exceeds 20 % of the initial baseline.

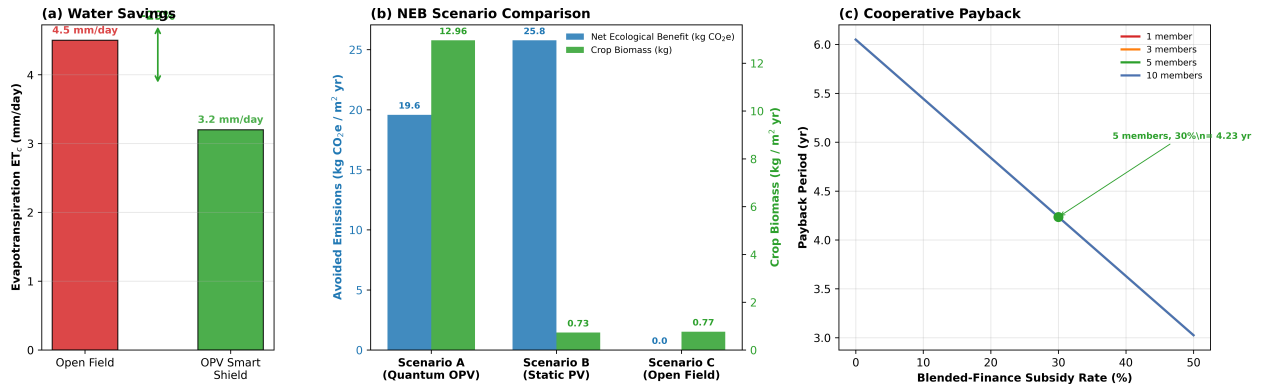


Figure 3: Water–Energy–Food network assessment. (a) FAO-56 evapotranspiration comparison: open field ($\text{ET}_c = 4.5 \text{ mm d}^{-1}$) vs. OPV smart shield ($\text{ET}_c = 3.2 \text{ mm d}^{-1}$), a 28 % reduction. (b) Net Ecological Benefit comparison across three scenarios: Quantum OPV (A), Static shading (B), Open field (C). Scenario A achieves $19.6 \text{ kg (CO}_2\text{)e/m}^2\text{/yr}$ with a crop biomass of $12.0 \text{ kg m}^{-2} \text{ yr}^{-1}$, while Scenario B reaches $25.8 \text{ kg (CO}_2\text{)e/m}^2\text{/yr}$ but only $0.68 \text{ kg m}^{-2} \text{ yr}^{-1}$ in biomass. The quantum shield preserves near-baseline crop productivity (0.971 global trapping yield) whereas static shading suppresses it to 0.064 , highlighting the yield–carbon trade-off central to the paradigm. The NEB serves as the verifiable metric for a tiered subsidy framework that bridges the quantum divide. (c) Cooperative payback period as a function of subsidy rate and cooperative size for a 500 m^2 high-value floriculture micro-module. For 5 members with 30 % blended-finance subsidy, payback = 4.23 yr (net annual cashflow of $26\,612 \text{ USD yr}^{-1}$ after dual OPEX). This validates that quantum-enhanced agrivoltaics can be economically transformative when coupled with cooperative ownership and strategic crop selection, mitigating the quantum divide for smallholder farmers.

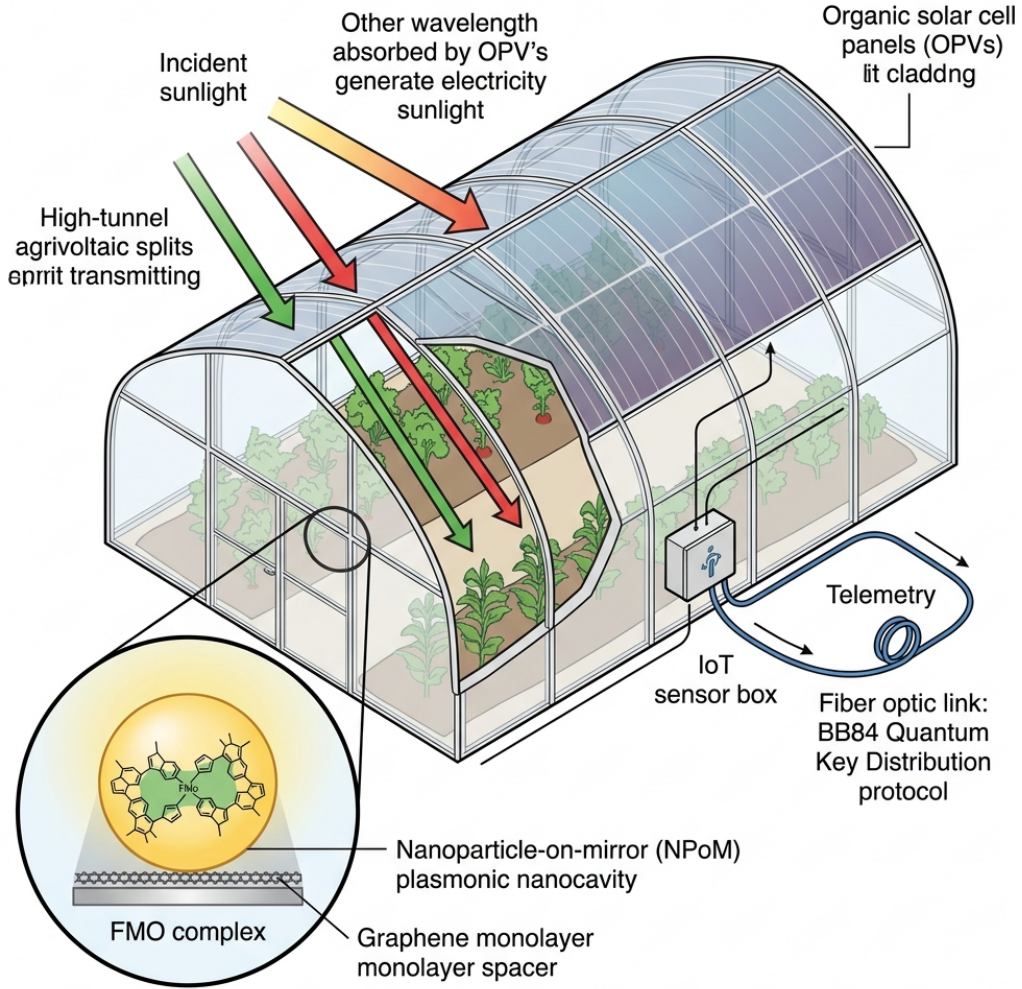


Figure 4: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server, securing agronomic actuator commands under a permissioned ledger architecture.

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